

CS:5980

Foundations of Embedded Systems

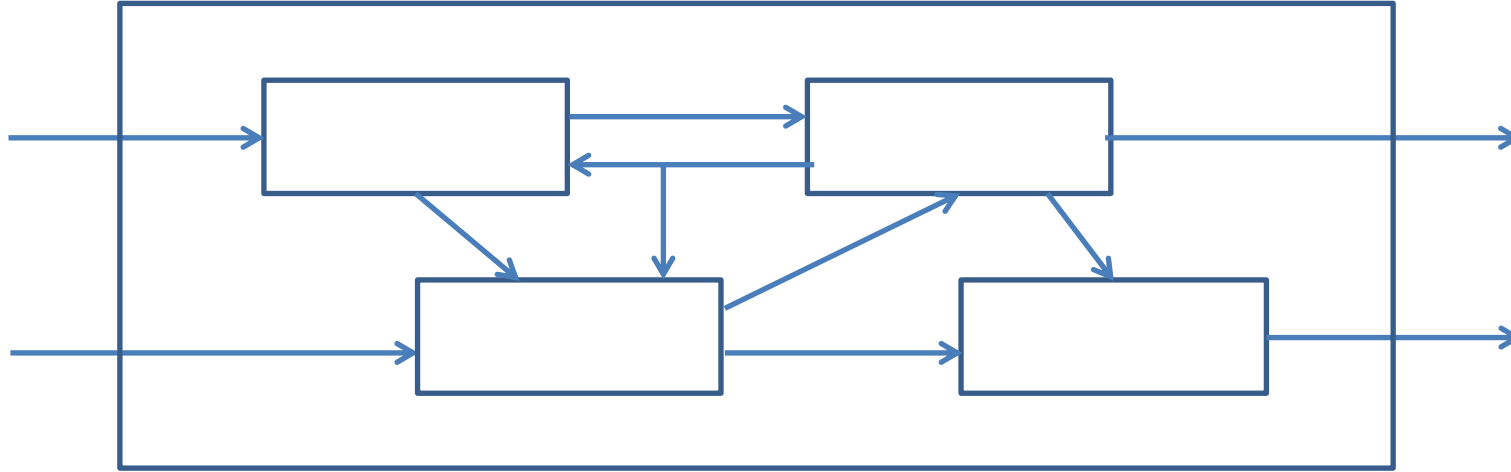
Synchronous Model

Part II

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Block Diagrams



Structured modeling

- How do we build complex models from simpler ones?
- What are basic operations on components?

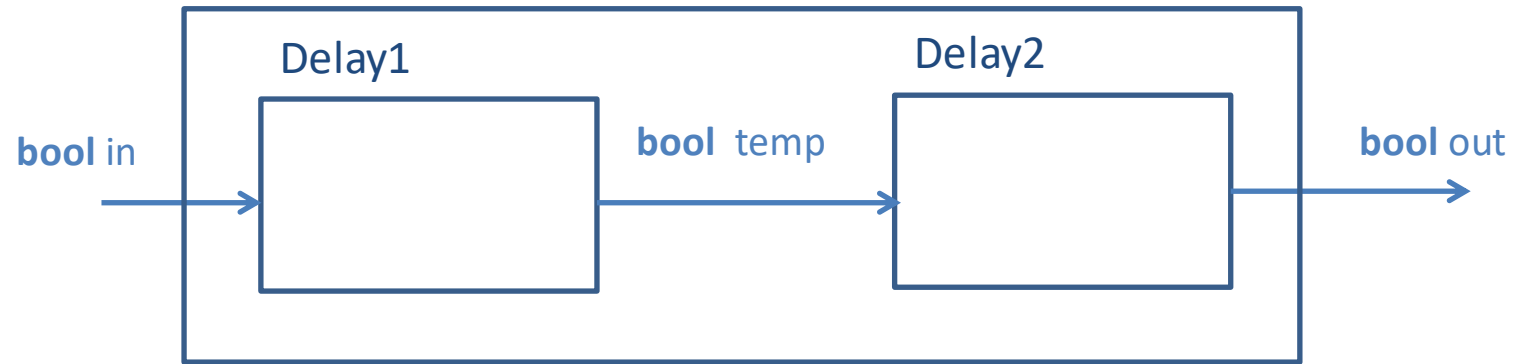
DoubleDelay



Design a component with

- **Input:** bool in
- **Output:** bool out
- **Behavior:** out in round n should equal in in round $n-2$

DoubleDelay



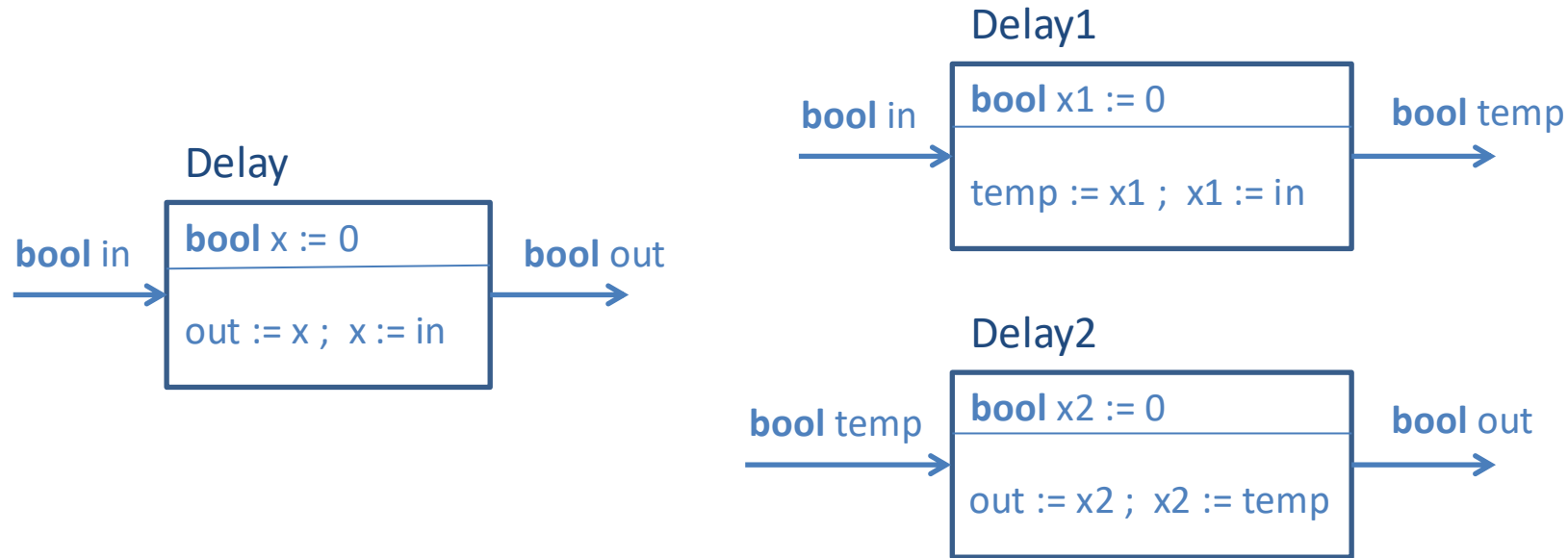
Instantiation: Create two instances of Delay

- output of Delay1 = input of Delay2 = variable temp

Parallel composition/product: Achieve concurrent execution of Delay1 and Delay2

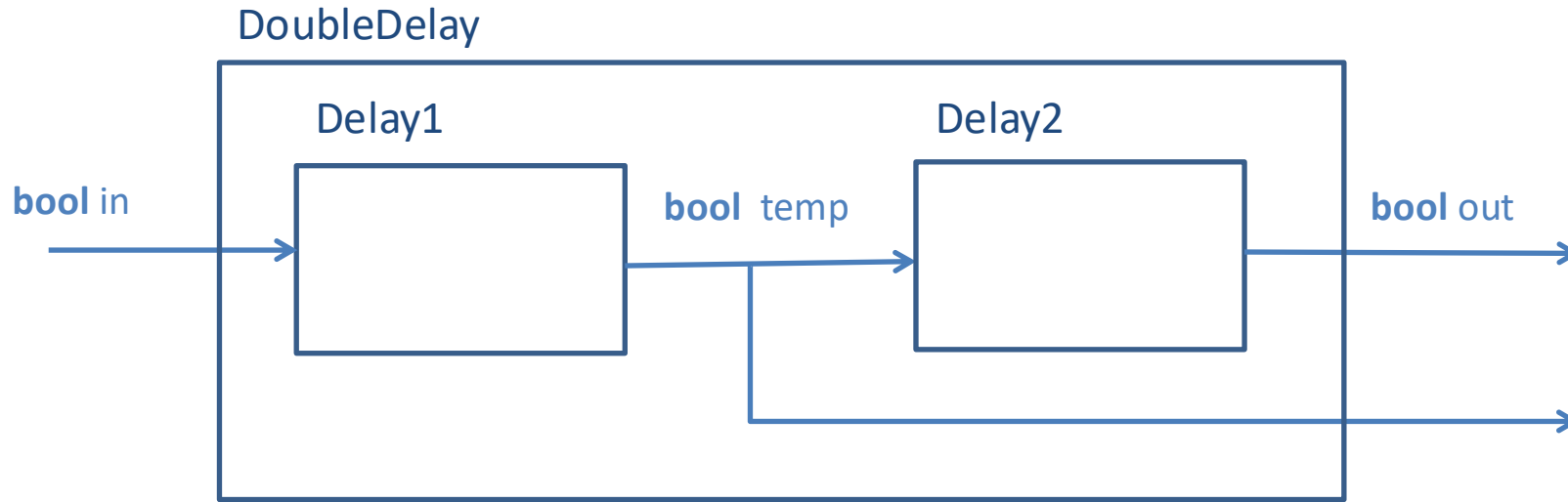
Encapsulation: Hide variable temp

Instantiation / Renaming



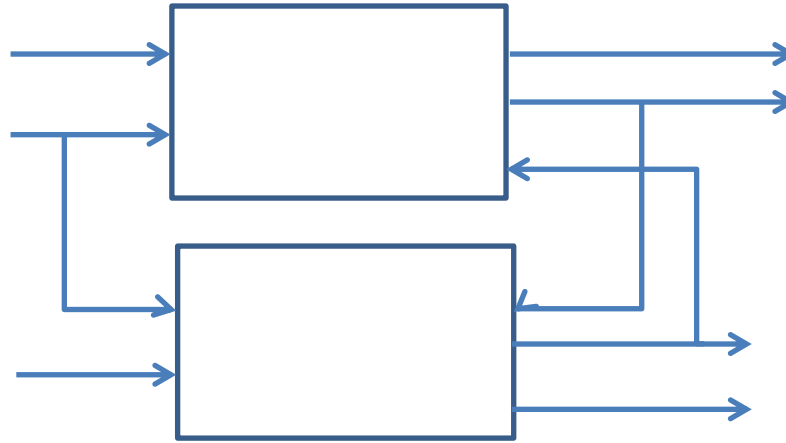
- $\text{Delay1} = \text{Delay}[\text{out} \mapsto \text{temp}]$
 - **Explicit** renaming of **input/output** variables
 - **Implicit** renaming apart of **state** variables
 - Definition $(I_1, O_1, S_1, \text{Init}_1, \text{React}_1)$ of **Delay1** derived from **Delay**
- $\text{Delay2} = \text{Delay}[\text{in} \mapsto \text{temp}]$

Parallel Composition (or Product)



- **DoubleDelay = Delay1 || Delay2**
 - Two components execute concurrently
- When can two components be composed?
- How to define parallel composition precisely?
 - Input/output/state variables, initialization, and reaction description of composite defined in terms of components
 - Can be viewed as an **algorithm** for compilation

Compatibility of Components C1 and C2



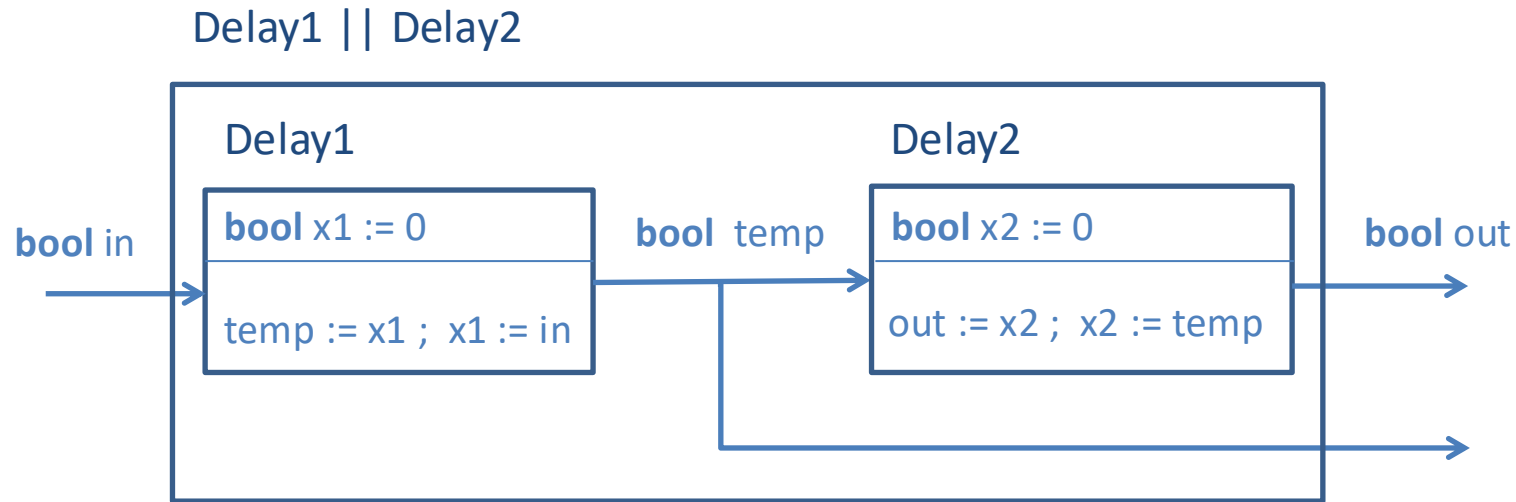
Allowed:

- Shared input variables
- Output variable of one is input variable of the other

Disallowed:

- Shared output variables
 - A unique component must be responsible for values of any given variable
- Shared state variables
 - However, state variables can always be renamed to avoid conflicts

Outputs of Product



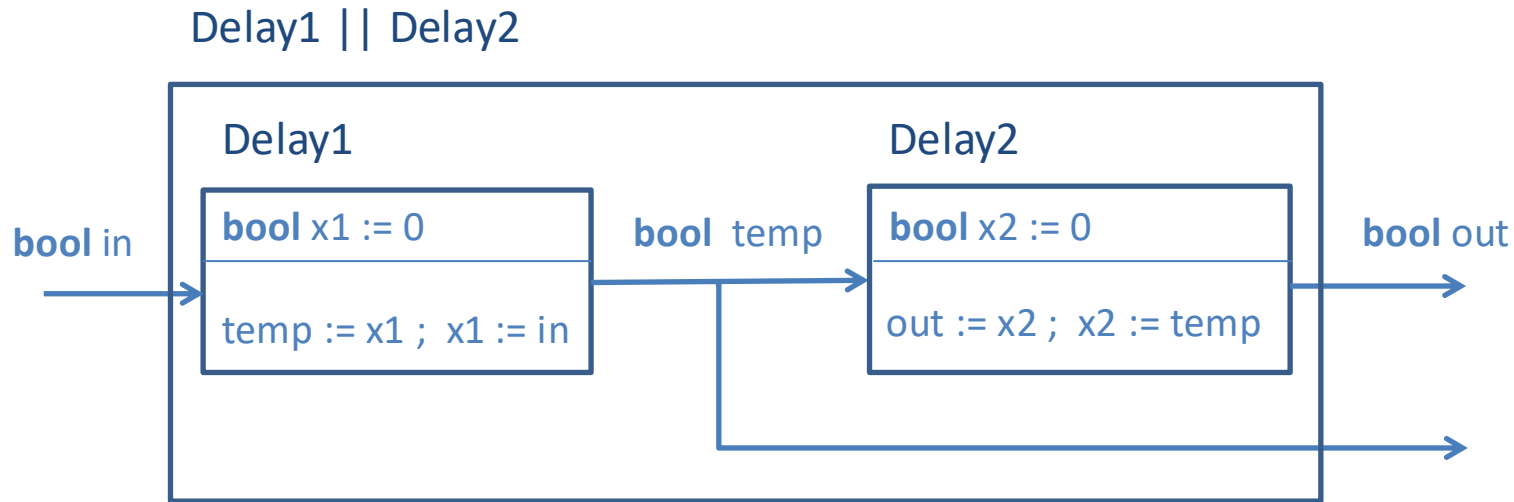
The **output variables** of $\text{Delay1} \parallel \text{Delay2}$ are $\{ \text{temp}, \text{out} \}$

- By default, every output of product is available to outside world

If C_1 has **output vars** O_1 and C_2 has **output vars** O_2

then $C_1 \parallel C_2$ has **output vars** $O_1 \cup O_2$

Inputs of Product



The **input variables** of Delay1 || Delay2 are { in }

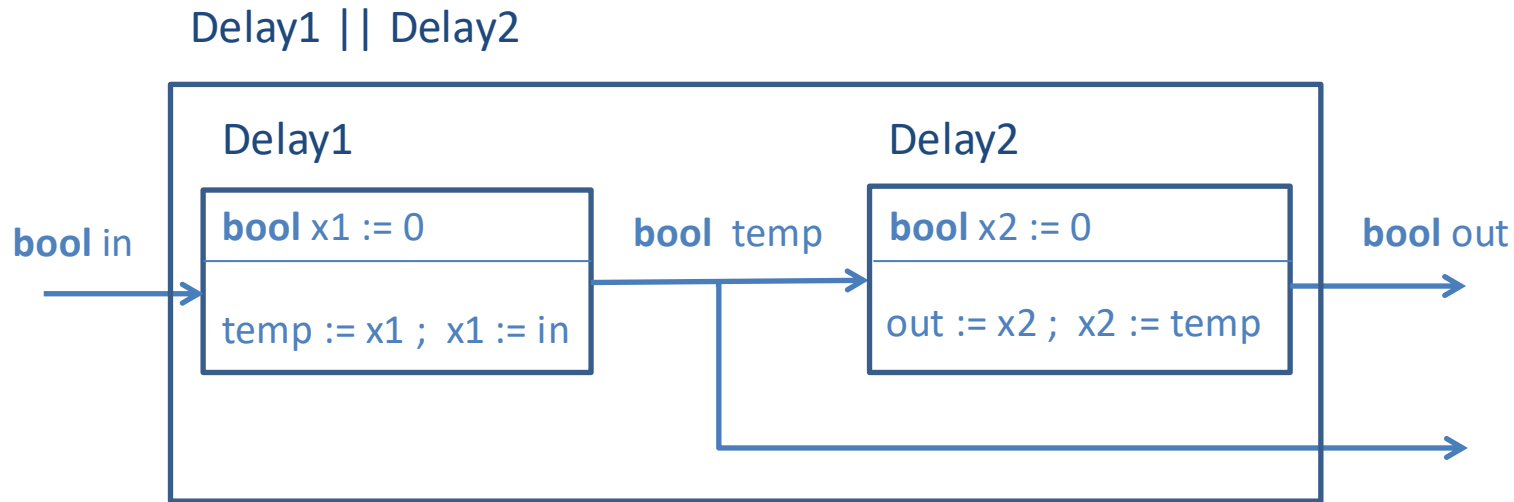
- Even though `temp` is input of Delay2, it is not an input of product

If C_1 has **input vars** I_1 and C_2 has **input vars** I_2

then $C_1 || C_2$ has **input vars** $(I_1 \cup I_2) \setminus (O_1 \cup O_2)$

- A variable is an input of the product iff
it is an input of one of the components, and not an output of the other

States of Product



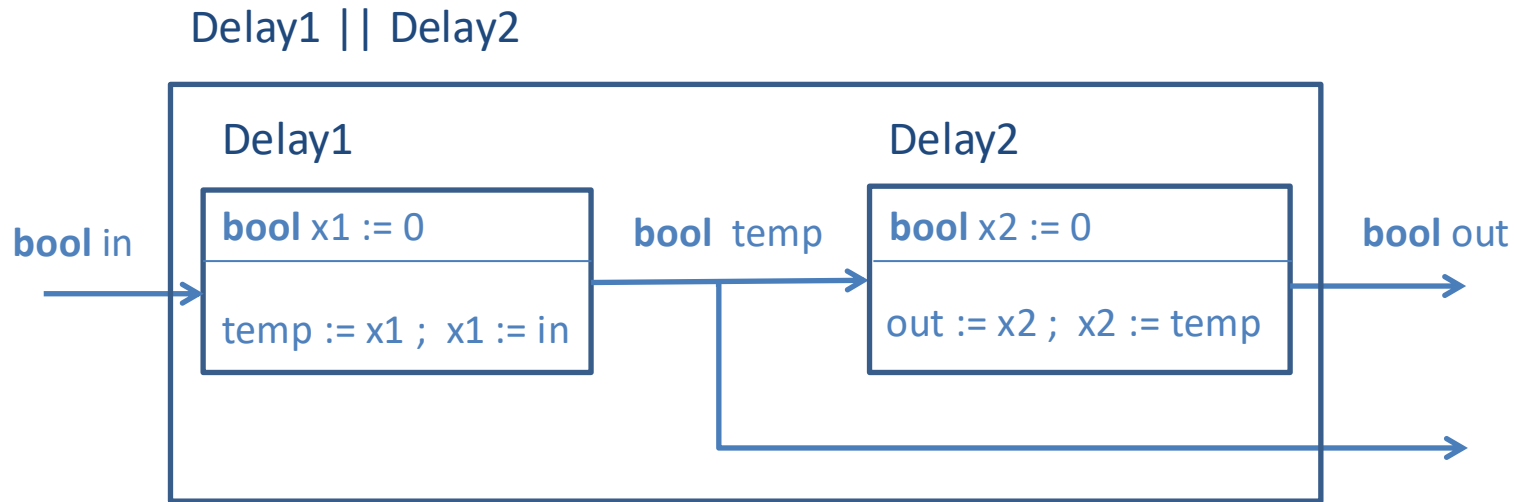
The **state variables** of Delay1 || Delay2 are $\{ x1, x2 \}$

If C_1 has **state vars** S_1 and C_2 has **state vars** S_2

then $C_1 || C_2$ has **state vars** $S_1 \cup S_2$ (recall that $S_1 \cap S_2 = \emptyset$)

- A state of the product is a pair (s_1, s_2) , where s_1 is a state of C_1 and s_2 is a state of C_2
- If C_1 has n_1 states and C_2 has n_2 states, then $C_1 || C_2$ has $n_1 \cdot n_2$ states

Initial States of Product



The **initialization code** Init for $\text{Delay1} \parallel \text{Delay2}$ is $x1 := 0 ; x2 := 0$

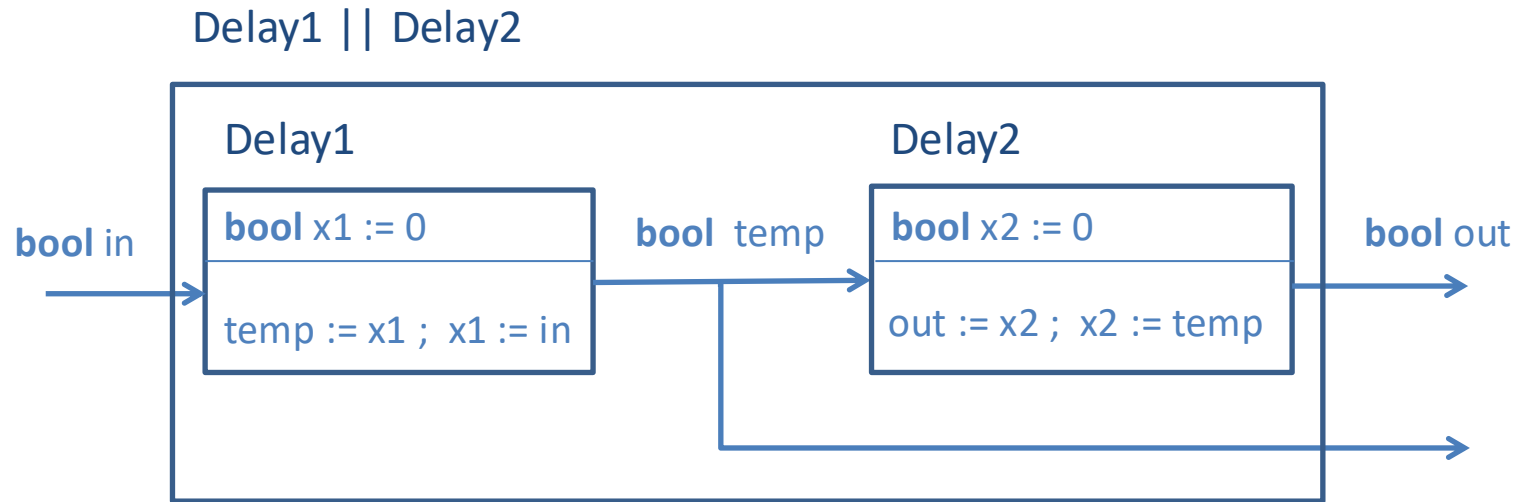
- Initial states are $\{ (0,0) \}$

If C_1 has **initialization** Init_1 and C_2 has **initialization** Init_2

then $C_1 \parallel C_2$ has **initialization** $\text{Init}_1 ; \text{Init}_2$ (or, equivalently, $\text{Init}_2 ; \text{Init}_1$)

- $\llbracket \text{Init} \rrbracket$ is the Cartesian product $\llbracket \text{Init}_1 \rrbracket \times \llbracket \text{Init}_2 \rrbracket$

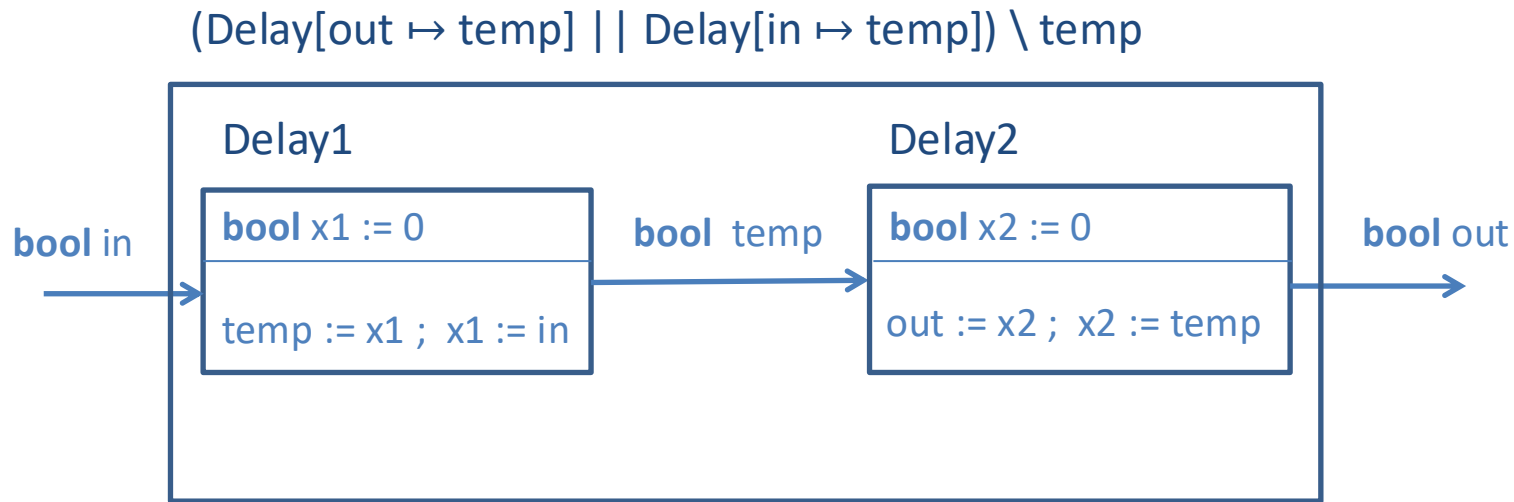
Reactions of Product



Execution of **Delay1 || Delay2** within a round:

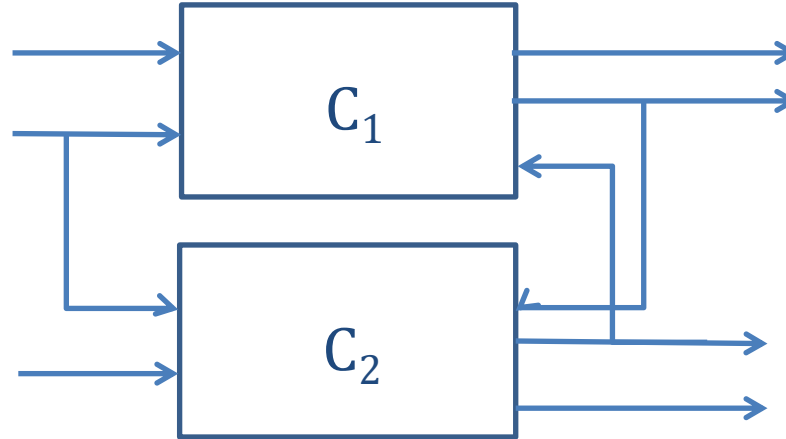
- Environment provides input value for variable **in**
- Concurrently execute code **temp := x1 ; x1 := in** of **Delay1**
- Concurrently execute code **out := x2 ; x2 := temp** of **Delay2**

Final Composition



- Instantiation: $\text{Delay}[\text{out} \mapsto \text{temp}]$ and $\text{Delay}[\text{in} \mapsto \text{temp}]$
- Parallel composition: $\text{Delay}[\text{out} \mapsto \text{temp}] \parallel \text{Delay}[\text{in} \mapsto \text{temp}]$
- Encapsulation: $(\text{Delay}[\text{out} \mapsto \text{temp}] \parallel \text{Delay}[\text{in} \mapsto \text{temp}]) \setminus \text{temp}$

Feedback Composition



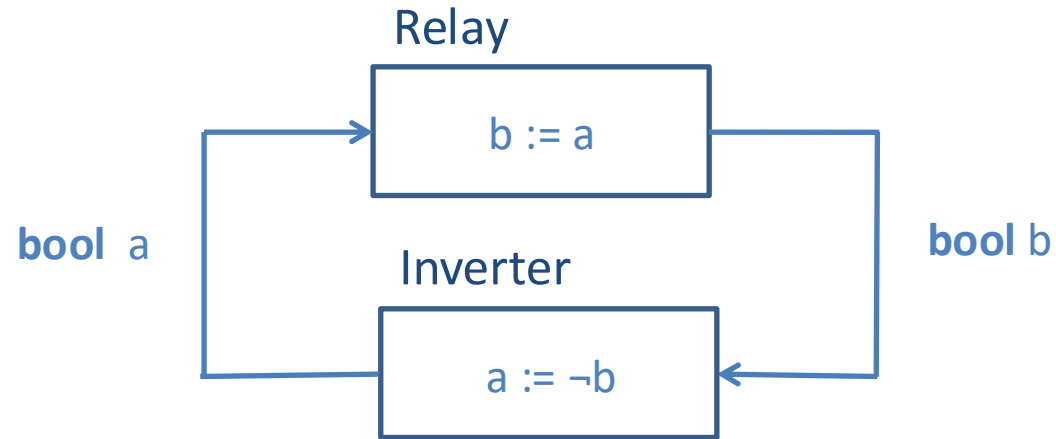
When

- some **output** of C_1 is an **input** of C_2 , and
- some **output** of C_2 is an **input** of C_1 ,

how do we order the executions of their reactions React_1 and React_2 ?

Should such composition be allowed at all?

Feedback Composition

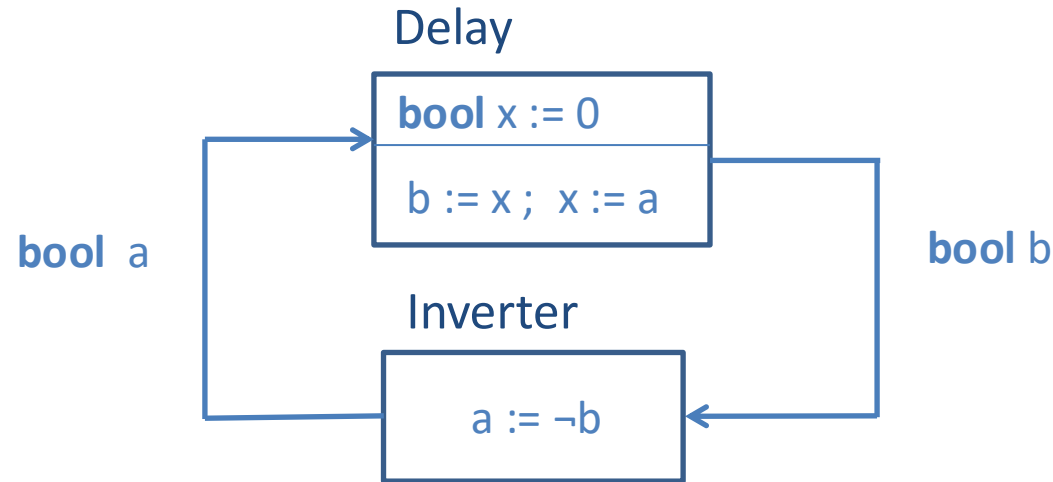


For **Relay**: its output **b** *awaits* its input **a**

For **Inverter**: its output **a** *awaits* its input **b**

- In product, we **cannot order** the execution of the two
- In the presence of such **cyclic dependency**, composition is **disallowed**
- **Intuition:** combinational **cycles** should be **avoided**

Feedback Composition



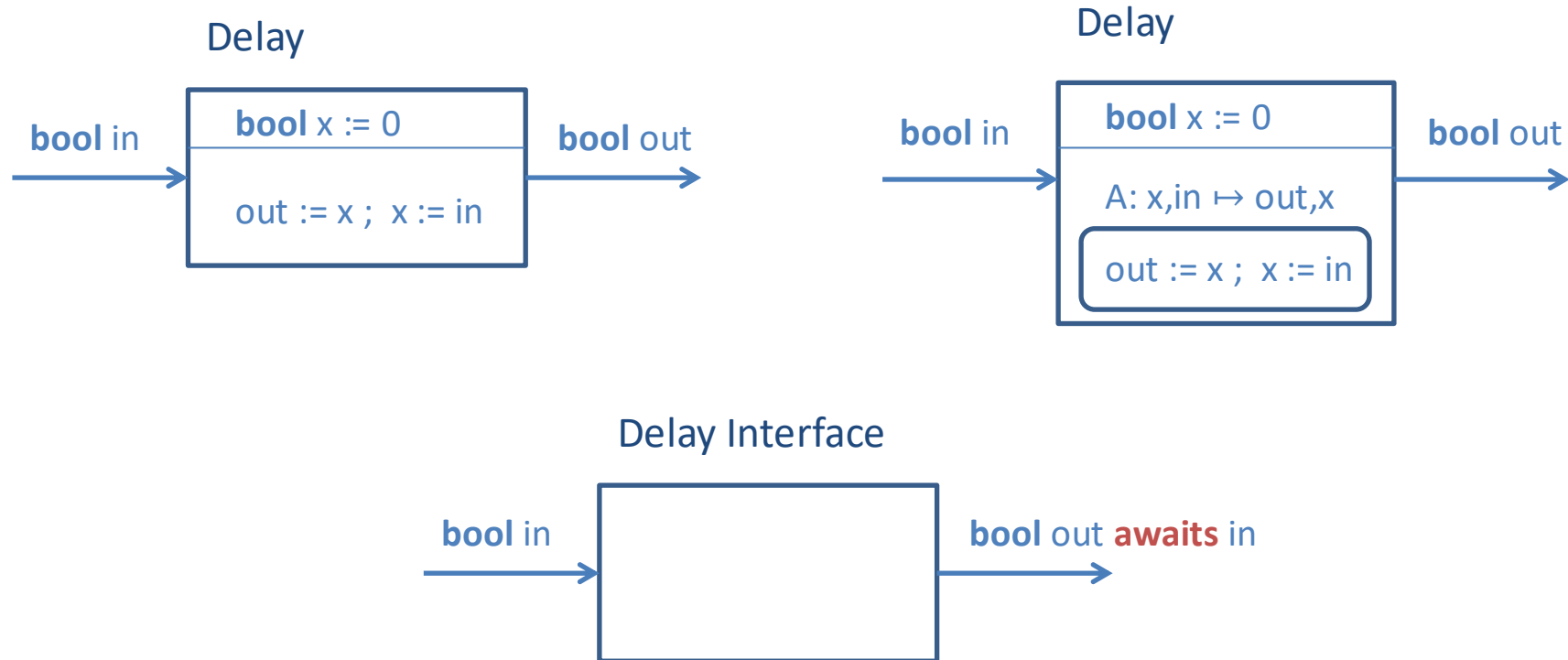
For **Delay**, output can be produced without waiting for input (by executing the assignment `b := x`)

Reaction code for **Delay || Inverter** could be `b := x ; a := ¬b ; x := a`

Goal: Refine specification of reaction description so that **await** dependencies among output-input vars are easy to detect

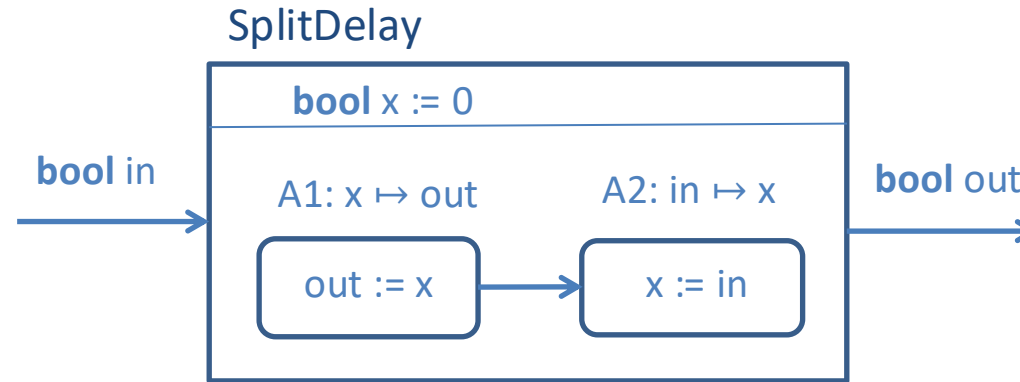
- Ordering of code-blocks during composition should be easy

Interfaces



Interface = (input variables, output variables, await dependencies)

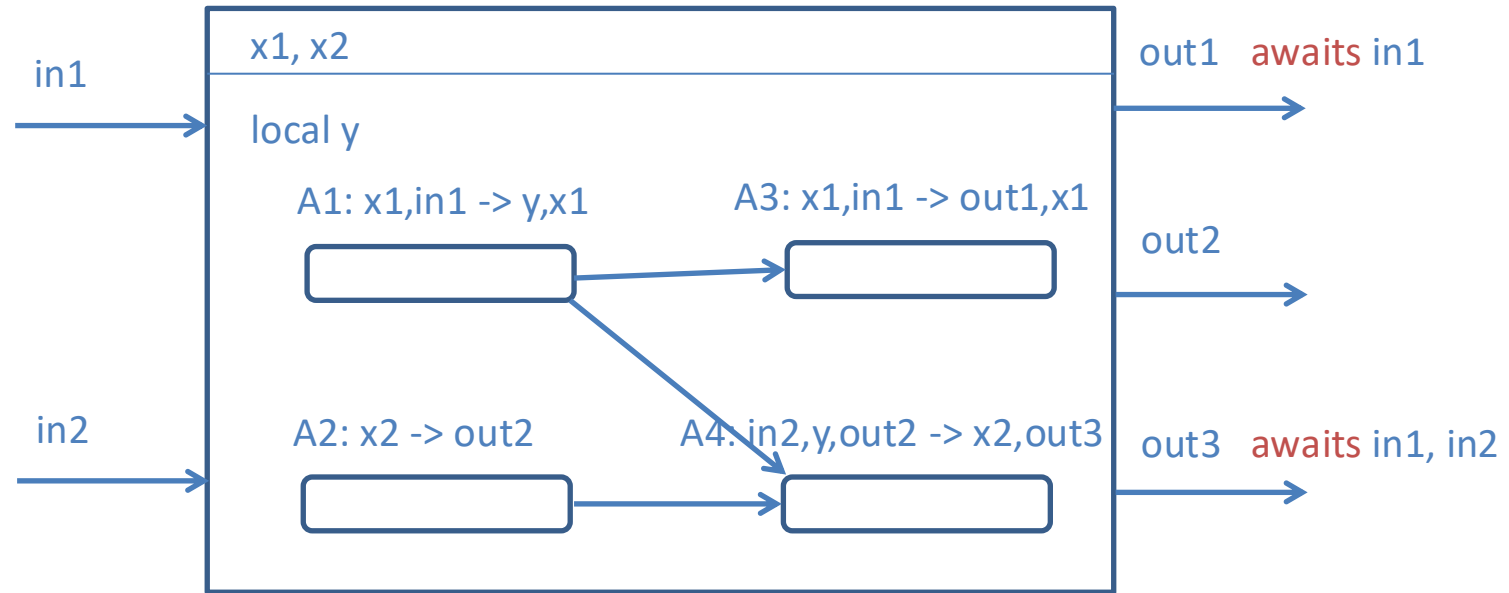
Interface: SplitDelay



Decomposing the reaction into tasks **eliminates** in this case the await dependency between **out** and **in**



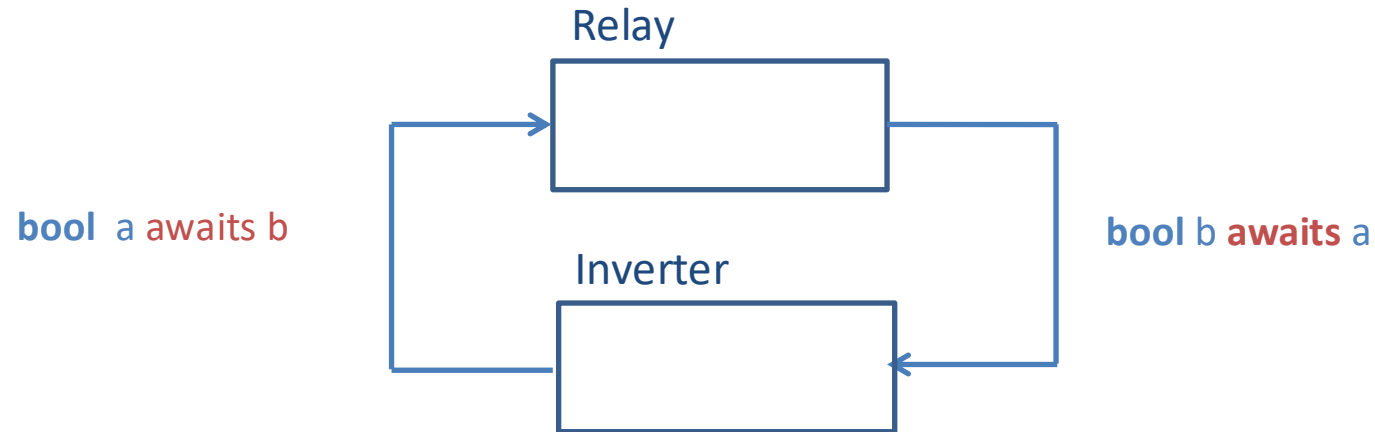
Example Interface



Example Interface



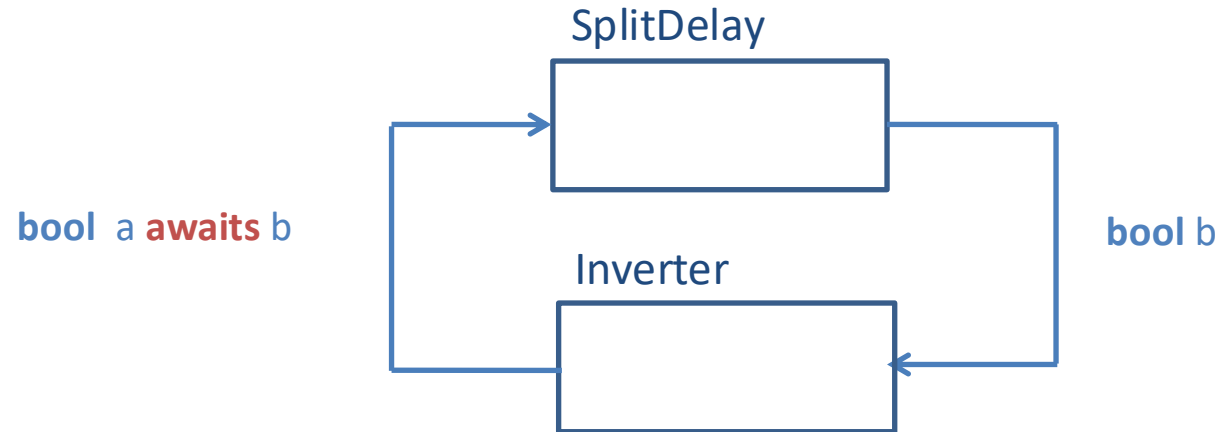
Back to Parallel Composition



Relay and Inverter are not compatible
since there is a cycle in their combined await dependencies

Note: We can conclude that based just on the components' interfaces

Composing SplitDelay and Inverter



SplitDelay and **Inverter** are compatible
since there is **no cycle** in their combined await dependencies

Component Compatibility Definition

Given components :

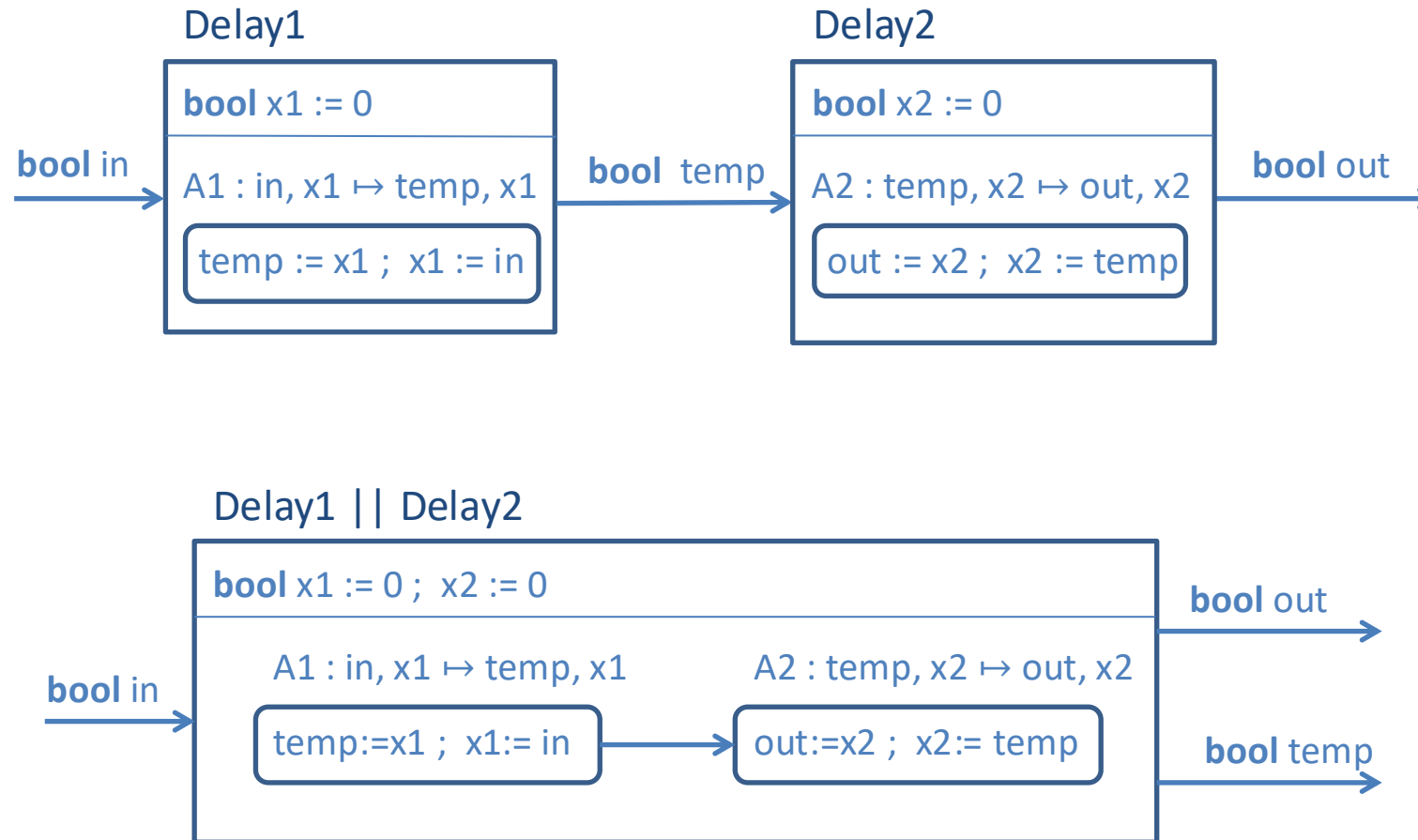
- C_1 with input vars I_1 , output vars O_1 , and awaits-dep. relation $\succ_1$
- C_2 with input vars I_2 , output vars O_2 , and awaits-dep. relation $\succ_2$

C_1 and C_2 are *compatible* if

1. they have no common outputs: sets O_1 and O_2 are disjoint
2. the relation $\succ_1 \cup \succ_2$ of combined await-dependencies is *acyclic*

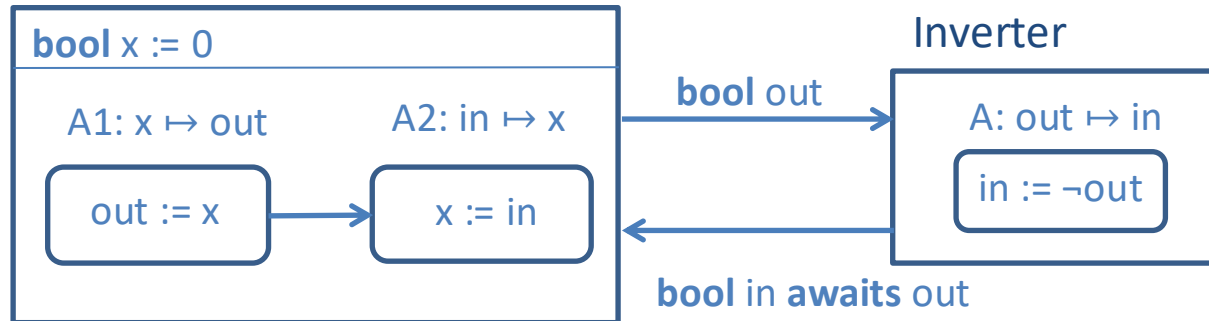
Parallel composition is *allowed only* for *compatible* components

Defining the Product

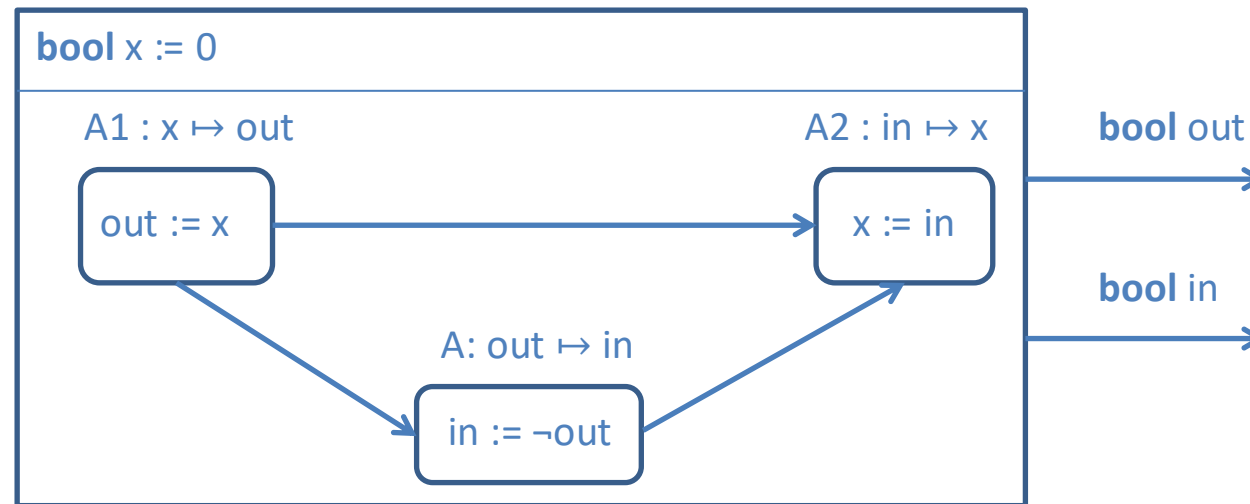


Composing SplitDelay and Inverter

SplitDelay



SplitDelay || Inverter



Parallel Composition Definition

Given compatible components

- $C_1 = (I_1, O_1, S_1, \text{Init}_1, \text{React}_1)$ and
- $C_2 = (I_2, O_2, S_2, \text{Init}_2, \text{React}_2),$

what's the reaction of product $C = C_1 \parallel C_2$?

Suppose React_1 and React_2 are specified using resp.

- local vars L_1 , set of tasks P_1 , and precedence $\prec_1$, and
- local vars L_2 , set of tasks P_2 , and precedence $\prec_2$

Reaction description for product C has

- local variables $L_1 \cup L_2$
- set of tasks $P_1 \cup P_2$
- precedence edges $\prec_1 \cup \prec_2 \cup \{ \text{edges between task } A_1 \text{ of } C_i \text{ and } A_2 \text{ of } C_j \mid i \neq j \text{ and } A_2 \text{ reads a var written by } A_1 \}$

Parallel Composition Definition

Why is the parallel composition operation well-defined?

- Can the new edges make task graph of the product cyclic?

Recall: Await-dependencies among I/O variables of compatible components must be acyclic

Proposition 2.1: Awaits compatibility implies acyclicity of product task graph

Bottom line: Interfaces capture enough information to define parallel composition in a consistent manner

Aside: It is possible to define more flexible (but more complex) notions of awaits dependency

Properties of Parallel Composition

Commutativity: $C_1 \parallel C_2 = C_2 \parallel C_1$ (when C_1, C_2 are compatible)

Associativity: $(C_1 \parallel C_2) \parallel C_3 = C_1 \parallel (C_2 \parallel C_3)$

- If compatibility check fails in one case, it will also fail in the others

Bottom line: order of composition does not matter

Note:

- If C_1 has n_1 states and C_2 has n_2 states, then $C_1 \parallel C_2$ has $n_1 \cdot n_2$ states
- If both C_1 and C_2 are deterministic, so is $C_1 \parallel C_2$
- If both C_1 and C_2 are event-triggered, is $C_1 \parallel C_2$ guaranteed to be event-triggered?

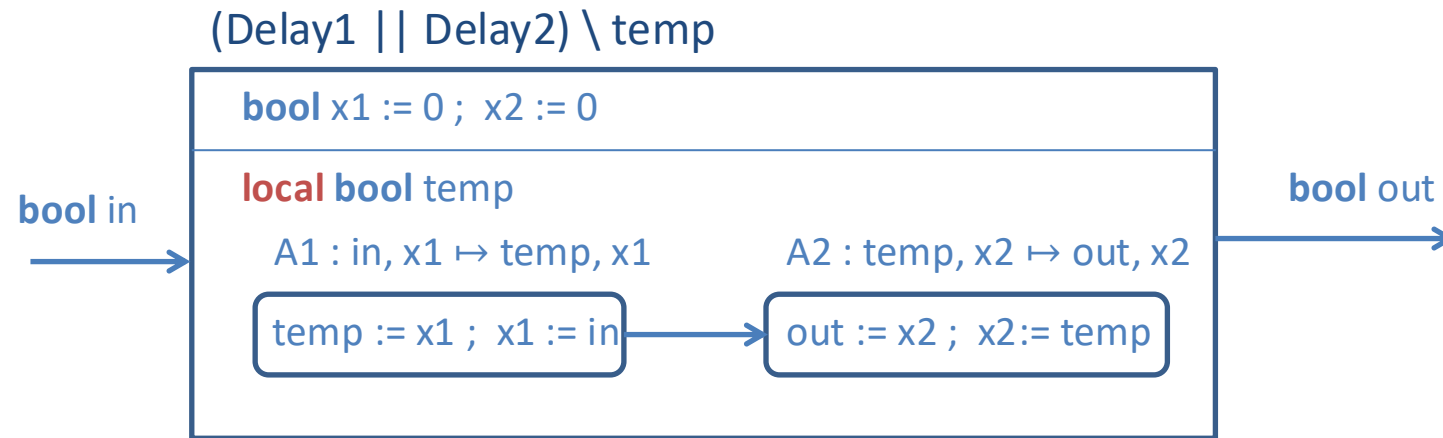
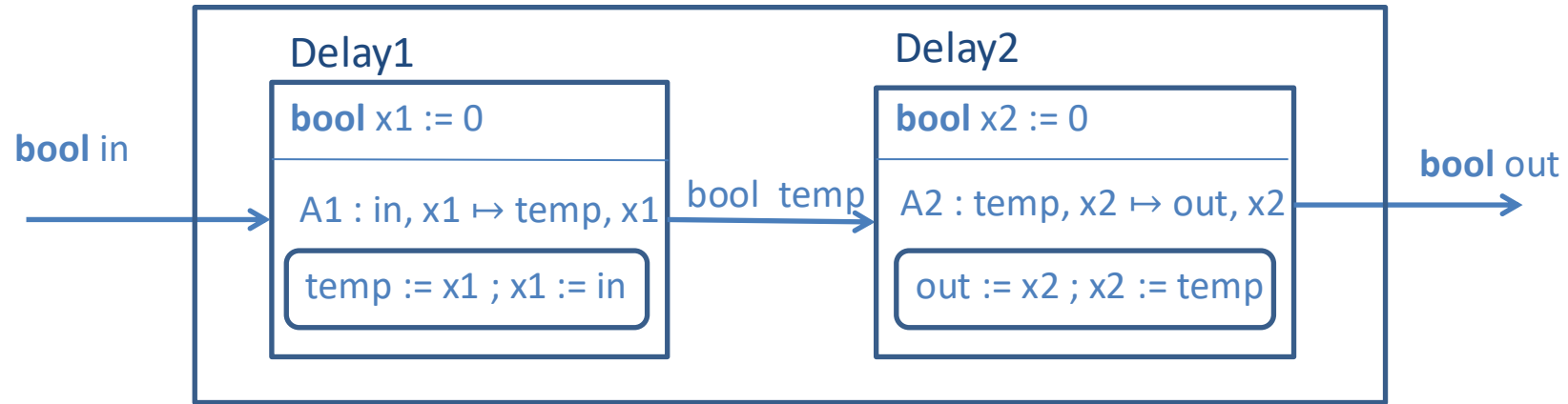
Output Hiding

Let C be a component and y one of its output vars

- The result of hiding y in C , written as $C \setminus y$, is a component identical to C except that y is a local variable

This is useful for limiting the scope of a component (encapsulation)

DoubleDelay



Credits

Notes based on Chapter 2 of

[Principles of Cyber-Physical Systems](#)

by Rajeev Alur

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